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Measuring responsible financial consumption behaviour

Measuring RFCB

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Sound consumption decisions determine individuals' well-being; responsible financial consumption behaviour affects not only their finances but also their societal status and emotional state. The failure to manage personal finances responsibly may result in serious long-term consequences for individuals and society overall. In order to evaluate the concept of responsible financial consumption behaviour, this study combines two established theoretical frameworks — the family management system and the theory of planned behaviour. The paper investigates the relationships among responsible financial consumption behaviour, responsible financial attitudes, financial literacy and behavioural control. Its theoretical model is tested on a random sample of 494 respondents and analysed

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using PLS-SEM. The results confirm the formation of responsible consumption behaviour by six formative elements: self-control in spending, planning for the future, seeking information, education, rational decision-making and solvency. The findings also indicate that all three focal variables have a direct effect on responsible financial consumption behaviour.

Keywords: responsible financial consumption behaviour, financial attitudes, financial literacy, behavioural control, family management system, theory of planned behaviour

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Introduction

Sound personal finance management has received increasing attention from the public and from academia. Consumers who make informed financial decision and behave in a financially responsible way are essential to stability and economic prosperity. In order to help consumers in improving financial decision-making, it is essential to deeply understand their financial behaviour (Van Raaij, 2016). Bad financial decisions presumably arise because people either lack critical information or the ability to understand and apply that information (García, 2013).

This view is in accordance with classical economic theory, which holds that informed consumers will demonstrate financially responsible behaviour (Persky, 1995; Thaler, 2000). Significant parts of economic and financial theory assume that individuals make financial decisions as "Homo economicus." In that stylized world, individuals behave with perfect rationality and make decisions by utilizing all available information within an efficient market. The relationship between financial knowledge and consumer behaviour is not as simple as the Homo economicus model assumes. If individuals truly acted rationally, markets would never need state interventions, since there would be no imbalances or bubbles.

The opposite stream of thinking developed at the end of the past century as psychologists and behavioural economists sought to understand human psychological phenomena and clarify individual financial behaviour. Diverse experiments established a new perspective of consumer behaviour within the field of behavioural economics and psychology. That approach is predominantly focused on the in-depth understanding of the effect of psychological obstacles on the attitude and knowledge behavioural gap. Irrational human behaviour and repeated, regular errors in financial decision-making challenge the existence of Homo economicus (Pompian, 2011).

The behavioural economics literature assumes that personal values and emotions significantly affect individuals' subjective perception of reality (Nga and Yien, 2013). Emotions make humans see the world from a different perspective (Ariely, 2009).

Kahneman and Tversky's (1979) investigation of human irrationality demonstrated that people make decisions in ways that predictably deviate from rationality and standard economic theory. There are two essential models of human information processing; an intuitive and speedy model (System 1) and an analytical and slow model (System 2) (Kahneman, 2011). The first system is very prone to cognitive bias and often leads to mistakes in decision-making; the other is more comprehensive and in charge of correcting the fast heuristics of first system (Kahneman and Frederick, 2005).

System 1 cannot be turned off; it operates automatically below the conscious intention and operates with minimal mental effort. System 2, on the other hand, allocates attention to deliberate mental activities, including analysis and complex computations (Chen, 2016).

This paper investigates the concept of responsible financial consumption behaviour and the factors that determine it. Variables that influence behaviour in general are knowledge and mental constructs such as attitudes, perception, control and motivation (Eagly and Chaiken, 1993; Van Raaij, 2016). To date, substantial research has emphasized the relationship

between financial literacy and financial behaviour. But very limited research has explored the connection between attitudes and sound financial behaviour.

To address the problem of responsible financial consumption behaviour, we question the assumptions of both classical and behavioural economics. This paper investigates financial literacy, responsible financial attitudes and behavioural control simultaneously in order to capture a holistic view of the psychological dimension of the financial behavioural gap arising from predictive irrationality. No other study has explored behavioural finance in relation to responsible financial consumption behaviour. The study extends previous studies and examines the consumption behaviour of individuals in detail. Enhanced consideration of the relationship between these variables will enable financial planners, educators, counsellors and program developers to strategically address the roots of overspending and poor management of consumption.

The framework for investigating responsible financial consumption behaviour

Theoretical background

Personal finance is a multidisciplinary field drawing upon family studies, economics, psychology, consumer behaviour and law. Two foundational theories support this study's conceptual model: (1) the family management system and (2) the theory of planned behaviour. Combining these two theories clarifies the connection between financial consumption behaviour, financial attitudes, financial knowledge, and behavioural control.

Inputs, throughputs and outputs present the main components of family management system model (Parrota and Johnson, 1998). When applied to the financial terminology inputs, throughputs and outputs stand for: financial knowledge, financial attitudes and financial management behaviour as shown in the Figure 1. Financial attitudes are defined as psychological tendencies expressed when a person evaluates recommended financial management practices. At the same time managerial subsystem is formed as a process of financial management comprising of planning, implementation and evaluation of financial tools.

The theory of planned behaviour is comprised of three types of reflections: behavioural, normative and control (Ajzen, 1991). Behavioural beliefs form affirmative or negative attitudes towards diverse objects. An increase in positive attitudes towards certain behaviour will increase the propensity to transform it into behaviour. Normative elements or also called subjective norms presents individuals' perception of social pressure; the way that

“thoughts and emotions of family, friends and colleagues” influence individual’s behaviour. Perceived behavioural control describes ones’ perception of own control over the situation: the behaviour that individual exhibits due to internal or external perception of obstacles (Ajzen, 1991). The interplay of the relevant contributors to the behaviour is portrayed in the Figure 2.

Defining responsible financial consumption behaviour

It is very hard to define what comprises “responsible financial consumption behaviour.” Xiao and Dew (2011) proposed a comprehensive measure of financial behaviour which included the following financial activities: cash management, savings and investments, credit use, insurance and consumption management. Xiao and Dew’s “Shopping and purchases” measure served as a benchmark in the development of responsible financial consumption behaviour measure used in this paper. We expand Xiao and Dew’s model to account for responsible decision-making in everyday activities, not just behaviour related to purely financial products and services.

Responsible financial consumption behaviour is connected to the psychological theories of Albert Maslow and the related field of Maslowian portfolio theory. According to Maslow, human needs are portrayed as layers of a pyramid; the bottom is formed by basic existential needs, followed by achievement, reputation, prestige and social standing finished on the top with self-actualization (Chen, 2016). Therefore, one way for individuals to achieve responsible consumption and financial progress is to satisfy their needs gradually, starting from the most basic human needs (survival) and then continuing to upper layers such as status, prestige, and self-actualization. Those who skip the basic layers and proceed immediately to self-gratification might be more susceptible to irresponsible consumption and poor financial performance.

For several reasons, this paper models responsible financial consumption behaviour as a “unidimensional” formative variable. First, behaviours related to savings, investments, insurance, and real estate have a profound impact upon financial well-being and security. Because it affects “decisions about what to buy, how to pay for” purchases, “and how ... [to] spread financial commitments over time,” “financial literacy is crucial” for personal as well as “social welfare” (Clark, 2014:2). Poor cash management and overconsumption can have detrimental consequences for individuals and households, and may result in over-indebtedness, bankruptcy, poverty, and financial and social exclusion.

Second, although not all people engage in saving, investing or insurance, virtually all individuals are consumers and, if nothing else, participate in behaviours that maximize their consumption benefits (Xiao and Dew, 2011).

Third, consumption precedes all other types of financial behaviours and actually represents one of the preconditions for almost all other kinds of behaviour. For example, a person cannot engage in investment behaviour if he or she has not created a budgetary surplus by refraining from consumption.

Finally, consumption practices significantly determine overall financial behaviour. For example, if financial consumption behaviour is poor and a person spends beyond his or her means, he or she will be unable to participate in saving, investing, or insurance (Kimball, 1990) and may even be forced to borrow in order to compensate for a budgetary shortfall.

The concept of responsible financial behaviour should be defined through the 6 designated dimensions as shown in the Figure 3.

Spending self-control

Conventional wisdom counsels individuals to think about the possible consequences of their actions. People should “think before they jump.” According to Kahneman’s theory (2011) of fast and slow thinking, individuals should learn and repeat certain actions until they become routine. People should move System 2 processes to System 1.

Self-control is seen as one’s ability to track, adjust and manage own behaviour according to the selected rules and standards with the clear capacity to enforce change (Carver and Scheier, 1998; Baumeister, 2002). The dominance of self-control over frivolous overspending has been proven as a relevant contributor to sound financial and consumption behaviour (Haws et al., 2012). Self-control is constant battle of consumption choices between current wants and the pressure of “irresistible impulses” and the determination to stick to the rationality of financial planning that consider not so immediate future (Wertenbroch, 1998; Baumeister, 2002; Haws et al., 2012).

Spending self-predictions and planning for the future

Personal spending predictions have profound practical implications. Everyday choices and important life decisions usually involve future expenditures. Because individuals’ predictions about themselves may lead to many consequential decisions and behaviours, psychologists have devoted significant effort to examining the success in the prediction of

future behaviour (Peetz and Buehler, 2016). The primary benefit of financial literacy, after all, is the ability “to make informed decisions that are consistent with [one’s] long-term interests” (Clark, 2014:4). The amount of effort one invests into monitoring its own behaviour is linked to the actual consumer choices; especially in controlling its impulsive purchases (Baumeister, 2002).

Seeking consumer information

In classical economics Adam Smith argued that informed consumers enforce vendors to be honest and play by the rules (Hogarth and Hilgert, 2003). The ability to comprehend the information on financial products and services is becoming even more crucial as the financial market is gaining its complexity. Complex financial products “have proven to be difficult for financially unsophisticated investors to master” (Lusardi and Mitchell, 2014:6).

Being informed means knowing where to find consumer information, how to use it, and how it can help. Research indicates that seeking information and reading literature can be an effective way to change individual behaviours (Gardner and Stern, 1996).

Taking part in consumer education

There is a clear difference between seeking information and education. Education usually involves a combination of information, knowledge, skill-development and motivation with a purpose of changing behaviour (Hilgert et al., 2003). According to Seitz (1972), consumer education is the process by which people comprehend market mechanisms so that they can improve their understanding of products and services they deem most likely to enhance their well-being.

Financial education and knowledge have a direct impact on favourable consumption and financial behaviour (Hilgert et al., 2003). Grønhaug (1974) demonstrated that education has some effect on the behaviour of the buyer. Compared to the less educated, consumers with a higher level of financial education should be more likely to follow recommended financial practices, such as paying bills on time, monitoring expenses every month and having an emergency fund (Hilgert et al., 2003). Furthermore, apart from the retirement planning education on specific type of financial products and services improves specific financial behaviour; such as credit card counselling (Elliehausen et al., 2003) or real-estate financing education programs (Mandell and Klein, 2009). Specific types of financial education improve specific financial behaviour; accordingly, consumer education should be included in the definition of responsible consumer behaviour.

Rational consumer decision-making

According to clinical psychology, a rational person should not engage in impulsive behaviour. Freud and his followers defined two basic human behavioural processes relative to the degree of impulsiveness; primary and secondary (Freud, 1911). The primary process promotes impulsive behaviour, while the secondary process promotes rationality. The relationship between primary and secondary thought processes is compatible with the concept of Systems 1 and 2 in behavioural economics.

Literature in clinical psychology and consumer finance alike emphasizes the correlation between impulsive acts and negative outcomes. Impulsive behaviour is almost always considered as bad, immature and irrational leading to decision-making errors (Hausman, 2000), whereas more rational behaviour arising from the secondary system is a complex, time-consuming process (Piron, 1991; Stern, 1962). Much of the psychological literature suggests that impulsive behaviour results in bad decisions, while a rational, analytical and more thorough approach leads to more responsible behaviour (Rook and Fisher, 1995).

From the rational point of view, impulsive behaviour and thoughtless buying seem wrong. Therefore, rational decision-making should contribute to the definition of responsible financial consumption behaviour.

Solvency

Solvency refers to a person's ability to pay his or her liabilities on time. Insolvent individuals and households could face a range of financial problems, since their liabilities exceed their assets (Garrett and James, 2013). According to Hogarth et al. (2003), the most basic practice which should contribute to successful financial behaviour is paying bills on time. Insolvency usually suggests that a person did not properly manage his or her spending. High levels of leverage might result from impulsive spending, poor cash-flow management or poor control over spending and consumption. Insolvency might imply irresponsible consumer financial consumption behaviour.

Development of research hypothesis

This research is based on the concept of responsible financial consumption behaviour and builds upon two theoretical models. There are seven hypotheses that examine the relationships among responsible financial consumption behaviour, personal attitudes, financial literacy and behavioural control shown in the Figure 4.

The literature suggests that both attitudes and knowledge contribute to personal finance management (Godwin, 1994). The relationship between financial attitudes determining financial behaviour has been proven numerous times (Hilgert and Hogarth, 2003). Individual money management behaviour such as spending or saving is formed by financial attitudes.

According to Atkinson and Messy (2012), attitudes are an essential component of financial literacy and financial behaviour. If people have a rather negative attitude towards a certain form of behaviour (such as saving) they will be less inclined to undertake such behaviour. Those believing money represents power are prone to engage in overspending or irresponsible consumption decisions (Tatzel, 2003). These propensities appear to prevail globally (Lemrová et al., 2014).

Excess levels of compulsive consumption are linked to financial attitudes (Roberts and Jones, 2001). According to Jorgensen et al. (2016), one potentially effective way to induce responsible spending is identifying and addressing individuals' financial attitudes related to spending and consumption processes.

Furthermore, significant change in behaviours is associated with a change in attitude (Xiao et al., 1995). A correlation between financial management and financial attitudes was established by Godwin (1994), who showed that attitudes towards planning are the greatest contributor to sound financial management practice.

H1: Responsible financial attitudes positively influence responsible consumer behaviour

Financial literacy comprises financial knowledge and financial skills necessary for theoretical and practical understanding of financial concepts and objects (Cvrlje, 2014). Financial literacy consists of “the ability to make informed judgments and ... effective decisions regarding the use and management of money” (Noctor et al., 1992). The general hypothesis is that increased financial literacy will improve financial attitudes and behaviour. Many studies have established a relationship between financial literacy and attitude (Ajzen, 1991) and financial literacy and behaviour (Hogarth et al., 2003).

The lack of knowledge of financial principles, and money management may explain why some individuals manage their resources inefficiently and fail to follow recommended financial practices (García, 2013). Individuals with a higher level of financial literacy should be able to make balanced financial decisions (Atkinson and Messy, 2012). Hilgert and Hogarth (2003) concluded that individuals who scored better on a financial knowledge test were likelier to follow recommended financial practices, such as making regular bill payments, and creating an emergency savings fund. A different study by Hilgert and others (2003) found that financially knowledgeable individuals are likelier to behave in financially responsible ways.

H2: Financial literacy positively influences responsible financial consumption behaviour

H5: Financial literacy positively influences responsible financial attitude

Unlike neoclassical economists, behavioural economists acknowledge the role of psychological characteristics and mental constructs in individual decision-making.

Many authors have analysed psychological factors and confirmed their significance in financial behaviour (Tokunaga, 1993; Hilgert et al., 2003; Hogarth et al., 2003; Atkinson and Messy, 2012; Cvrlje, 2014). Hira and Mugenda (1999) stressed the equal weighting of psychological factors alongside financial and practical factors. Materialism, impulsivity, self-efficacy, optimism, behavioural control, risk tolerance, financial planning horizon and attitudes may explain an individual's beliefs and behaviour with respect to money (Yamauchi and Templer, 1982). However, "irresistible impulses" to consume often lead to a sense of helplessness, and eventually towards irrational overspending (Rook, 1987; Baumeister, 2002).

Rubinstein (1981) argued that general attitudes and feelings towards life predominantly form their money mind set. Behavioural control may be defined as an individual's perception of his or her ability to perform a given task. It refers to the extent to which a person believes that he or she has the skills, resources, and other prerequisites to perform a given task (Ajzen, 1991).

Ajzen (1991) suggested that behavioural control affects individual's attitudes. There is a positive relationship between external locus of control and credit card debt accumulation (Davies and Lea, 1995). At the same time, responsible financial behaviour in terms of saving habits, budgeting and control is dependent on one's perception of control over life and finances (Perry and Morris, 2005).

H3: High level of behavioural control positively influences responsible financial consumption behaviour

H4: High level of behavioural control positively influences responsible financial attitudes

According to the family resource management system (Parrotta and Johnson, 1998), attitudes mediate the relationship between financial knowledge and financial behaviour. Consequently, responsible financial attitudes mediate the relationship between financial literacy and responsible financial consumption behaviour.

H6: The relationship between financial literacy and responsible financial consumption behaviour is mediated by responsible financial attitudes

Most of the relevant psychological literature suggests that the attitude-behaviour relationship might be affected by a third variable, such as individuals' knowledge and skills. Eagly and Chaiken (1993) suggested that the relationship between financial attitudes and behaviour is moderated by knowledge.

Parrotta and Johnson (1998) investigated the effects of financial knowledge on the relationship between financial attitudes and financial management concluding that there is no moderation between attitudes and behaviour. This assumption is tested by using a model derived from Parrotta and Johnson's (1998) adaptation of the family resource management system.

H7: Financial literacy moderates the relationship between responsible financial attitudes and responsible financial consumption behaviour

Methodology

Measurement scales

Financial literacy (FL) is modeled as a formative variable based on two indices: financial knowledge and financial skills. The concept itself does not have a generally accepted definition. According to Cvrlje (2014) should be investigated as a combination of

financial knowledge and financial skills. Many authors have agreed that financial knowledge is the most important element of financial literacy (Huston, 2010; Atkinson and Messy, 2012). Financial knowledge represents theoretical and practical understanding of financial concepts and facts. Financial skills represent the practical component of financial literacy. Financial skills include the ability to use acquired financial knowledge as well as self-confidence, which is needed to transform certain capabilities into actions (Huston, 2010). Items of the financial literacy scale were tested using the nominal scale (correct, incorrect and I do not know) that was developed by Lusardi and Mitchell (2007) and Crvlje (2014) and is used in a wide variety of research. Skills were tested with three items such as “If the annual interest rate on savings is 1% and annual inflation rate 2%, after a year I will have more money than today.” and knowledge with four similar to “Bonds are riskier than stocks”.

Financial attitudes were measured by three statements using the five-point Likert-type scale derived from the Theory of planned behaviour (Ajzen 2002, 2006). The scale measures personal attitudes towards responsible financial management: „I consider myself qualified to manage my financial situation and financial decision-making“; „I think it is extremely important to create a financial plan for the future.“; „I think it is extremely important to adhere to personal financial budget.“ Higher values should imply greater receptivity towards responsible consumer behaviour.

Individuals’ behavioural control was assessed using three items developed by Nysveen and others (2005): “I feel free to make my own financial decisions“; „My financial situation is entirely within my control“ and „I have the necessary resources and means to achieve financial success.“ Using the five-point Likert-type scale variable measures perceived control over the performance of financial tasks.

The purpose of this research was to form and to test the new formative variable – the responsible financial consumption behaviour. In the case of responsible financial consumption behaviour, causative indicators drive latent variables and not vice versa (MacCallum and Browne, 1993). Such measures do not follow the conventional procedure for scale development of new constructs (Churchill, 1979). Six items capture the entire context of the chosen construct. The extensive exploratory research was conducted that was in line with published standard guidelines (De Vellis, 1991). Moreover, to align the process of index formation with proposed procedure, we defined and tested the construct, object and attributes according to C-OAR-SE procedure (Rossiter, 2002). Such an approach is adequate for the development of a formative constructs scale. Upon completing the six proposed steps

of C-OAR-SE procedure (Rossiter, 2002), we defined the construct as responsible financial consumption behaviour as perceived by consumers with 6 dimensions and each is tested with single item “I live in accordance with my financial possibilities”, “I pay my monthly bills before on their due date”, “I try to anticipate and plan future financial income and expenses”, “In everyday purchases of consumer goods, I am informed to make rational decisions that are in line with my financial possibilities”, “I try to be informed and educated in the field of successful management of personal finances” and “When buying an item, the first think I think about is whether I can afford it”.

Data collection

The quantitative research is based on a highly structured questionnaire administered in July 2014 to a stratified random sample in Croatia. The sample was stratified by age, gender and geographic position according to the structures known from Population census 2011, where complete or approximate proportionality was achieved. The results of this study may therefore be generalized on overall population.

The survey included the sample of respondents in Croatia aged 20-79. Respondents were contacted by telephone, using the base of randomly selected telephone numbers of both mobile and fixed numbers. According to the formula developed by Krejcie and Morgan (1970), statistical reliability of 95% with a 5% acceptable level of error, required gathering of at least 384 completely fully answered questionnaires to be able to generalize the results to the overall population of 3.219.580. A total of 640 respondents took part in the survey. These respondents generated 494 complete questionnaires.

Initial sample analysis showed that 56,9% of the respondents were female and 39,9% were male. The largest percentage of respondents was “mature,” from 20 to 54 years old (84,6%), followed by “old” 55 to 95 years old (14,5%). Most of the respondents lived in northwest Croatia (41%), followed by Adriatic Croatia (38%) and Pannonian Croatia (21%). The academic attainment of the respondents was relatively high: 4,9% had a master’s degree or higher; 42,5% had a bachelor’s degree; 45,3% had a high school diploma, only 7,3% had less than a secondary level of education.

The average net income per household was 9.656,10 HRK. (At the rate of 1,00 HRK = 0,133686 EUR, the average household income in the survey was equivalent to €1.290,89.) The average net income per household member was 2.700,10 HRK (€360,97).

Within the analysed sample, 94,1% of respondents were either employed or actively seeking employment. The remaining 5,9% were retired. The largest cohort of respondents

consisted of those employed for an indefinite period of time (67,6%). Another 18,30% were employed for a definite period of time; 3,8% were seasonally employed, and 6,9% were unemployed.

The results of the model estimation

This paper's model is tested under the multivariate PLS-SEM method, which is adequate for exploratory research testing new hypotheses (Hensler et al., 2009). The model consists of two reflective variables (control and attitudes) and two formative variables (responsible consumer behaviour and financial literacy). Using a two-step procedure, the model also tests whether financial literacy moderates the relationship between attitudes and responsible consumer behaviour. Finally, the model tests the mediating effect of control between financial literacy and responsible consumer behaviour.

Before the discussion of the significance of selected relationships between variables, the reliability and validity of the inner and the outer model must be analysed. The reflexive constructs are evaluated through the AVE, composite reliability of indicators and discriminant validity (Hensler et al., 2009). Average variance extracted (AVE) as the measure of convergent validity should be greater than 0,5 (Hair et al., 2014:103). Composite reliability (CR) takes into account variability of item loadings that should be greater than 0,7 and it is interpreted as the traditional indicator Cronbach Alpha (Hensler et al., 2009). The value of R^2 is defined as the coefficient of determination. According to Guo and others (2011:219-221), everything above 0,1 or 10% has interpretive significance and therefore can be used in further analysis. Both reflexive variables in the model satisfy all of the given criteria as it is portrayed in the Table 1. Responsible financial behaviour is strongly determined with *R-squared* of 0,557 and attitudes on average with 0,211 (Chin, 1998:323). Furthermore, the statistical significance of structural coefficients factor loadings of reflective variables is confirmed.

The first step in the analysis of convergent validity is the factor loading. To be valid for the predominant items in the model, the factor loading should be greater than 0,7 (Hair et al., 2014:103). Loadings from 0,4 to 0,7 should be tested and taken out of the model only if such move would improve AVE or CR (Hair et al., 2014:103). Tests for all items with smaller

loadings provided no significant improvement in the model's fit. They were therefore included.

The discriminant validity (Fornell and Lacker criteria) is also proven since the AVE of each variable is much larger than its squared correlations with other variables in the model (Attitudes-Control squared — 0,194; AVE for Attitudes 0,552; AVE for Control 0,663).

The model consists of two formative variables: responsible consumer behaviour, which is comprised of six relevant indicators, and financial literacy, which consists of knowledge and skills. Both knowledge and skills are indices formed by three and four items. Nomological validity has been proved throughout the evaluation of the broader theory and the causes and effects that we have demonstrated in the theoretical foundation of this paper. The significance of each indicator weight is ensured at $\alpha = 10\%$.

In addition to assessing each variable's validity and reliability, the adequacy of the entire model must be assessed. To ensure the validity of the model, Cohen's coefficient f^2 (1998; Hensler et al., 2009). Analysis of the coefficients indicates that financial literacy has an almost insignificant effect, control has a small effect and attitudes have a medium to strong effect on responsible consumer behaviour. With respect to the attitudes variable, control has a moderate effect, and financial literacy has a small effect. According to the Stone-Geisser test (Hensler et al., 2009) all of the values above suggest relevant predictive validity, Q^2 being > 0 for all of the variables in the model (RCB=0,482, Attitudes=0,139, Control=0,330, FL=0,719).

For all of these reasons, we conclude that the model is adequate for the analysis of structural equations between latent variables

Hypothesis testing

Structural coefficients among exogenous variables are interpreted as beta coefficients in standard regression. They must satisfy two criteria. First, they must have the theoretically predicted sign. Second, they must have statistical significance (Hensler et al., 2009). If only one of the mentioned criteria is satisfied, the hypothesis can be only partially accepted. In terms of the hypotheses developed, all of the hypotheses are supported empirically; Table 2 provides the summary of the results.

Mediation

Analysis of the mediation effect of financial attitudes on financial literacy and responsible financial behaviour reveals a significant indirect relationship. When there is a significant indirect effect, we can conclude there is mediation. The final step is to determine the strength of the mediation result through variance accounted for (VAF). In this case, VAF accounts for 57,5%. Based on VAF, control partially mediates the relationship between financial literacy and responsible financial consumption behaviour as it can be seen in Table 3. Fifty-seven percent of the effect of that relationship is explained by financial attitudes.

Moderation

The last hypothesis in the model concerns the moderating effect of financial literacy on the relationship between attitudes and responsible consumer behaviour.

The hypothesis tests the level at which the strength of the relationship between attitudes and RCB is a function of financial literacy. The moderation effect was tested to determine whether the association between attitudes and responsible consumer behaviour depends on financial literacy (modelled as combination of knowledge and skills). The moderating (interaction) effect accounts for the coefficient of -0.0850 (t value 1.275***; *** $p < 0.01$) with $R^2 = 0.564$.

In addition, the statistically significant standardized coefficient between attitudes and responsible consumer behaviour is 0,622. Between financial literacy and responsible consumer behaviour, the coefficient is 0,069. The results show a statistically significant moderating (interaction) effect, therefore we can conclude hypothesis 7 is supported as well. The relationship between attitudes and reasonable financial behaviour weakens as the levels of financial literacy increase.

Discussion

This study represents a first attempt to develop an index of responsible financial consumption behaviour (RFCB) using a nationally representative sample. The formatted index has been proven statistically and methodologically appropriate.

Another finding of this study concerns the analysis of the relationships among responsible financial consumption behaviour and three factors: financial literacy and two other factors representing mental constructs related to human behaviour, responsible financial

attitude and behavioural control. The evaluation of the relationships among these factors rested mainly on two schools of economic thought (neoclassical and behavioural economics) and two foundational theories: the family management system and the theory of planned behaviour. Moreover, according to Maslowian theory, responsible financial consumption behaviour means progressing systematically from survival to self-actualization.

Support for the hypotheses indicated that financial attitudes, financial literacy and behavioural control play an important role in explaining responsible financial consumption behaviour. All three variables were significantly and positively related to responsible financial consumption behaviour. Attitude turn out to be the most significant factor, followed by behavioural control and finally by financial literacy. The results confirm the theoretical background that in order to achieve responsible financial consumption behaviour one must have plausible attitudes (Atkinskon and Messy, 2012; Godwin, 1994; Hilgert and Hogarth, 2003; Roberts and Jones, 2001; Xiao et al., 1995). The contribution of financial literacy is relatively low but proven, in line with published research Atkinson and Messy, 2012; García, 2013; Hilgert et al., 2003; Hogarth et al., 2003). Financially knowledgeable individuals have to have positive attitudes and certain level of behavioural control in order to be successful in consumption responsibility. Behavioural control also have significant impact on the formation of confident attitudes proving that the two the most relevant contributor to responsible financial consumption behaviour are psychological characteristics and mental constructs (Ajzen, 1991; Davies and Lea, 1995; Perry and Morris, 2005). These findings also support the theory of planned behaviour and the family management system.

Analysis of the mediation effect of financial attitudes between financial literacy and responsible financial behaviour revealed a significant indirect relationship. The findings suggest that financial attitudes partially mediate the relationship between financial literacy and responsible financial consumption behaviour. These results are in line with Parrota and Johnson (1998).

Finally, the results of moderation analysis showed that financial literacy has a moderating role in the relationship between responsible financial attitudes and responsible financial consumption behaviour, confirming Parrota and Johnson (1998).

Even though the results confirm the hypothesis on the moderating role of financial knowledge, the overall findings suggest some unexpected facts. Most of the literature investigating moderation effects implies that when financial literacy level is high, the positive relationship between financial attitudes and behaviour will be stronger than when financial literacy is low. However, the findings showed that the moderation effect of financial literacy

is negative. An increase in financial literacy will cause the relationship between financial attitudes and behaviour to become weaker than when financial literacy level is low being in conflict with other studies (Eagly and Chaiken, 1993). The explanation for the discrepancy might arise from Kahneman's (2011) distinction between fast and slow thinking. When financial literacy is high, people tend to make their consumer decisions more according to "what they know" — their "knowledge and skills" — and less according to "what their guts tell them." Thanks to higher literacy, System 2 has the last word in their actions. In addition, some of their financial skills may already have become a part of their routine behaviour, or System 1.

On the other hand, when financial literacy is low, consumers' knowledge and skills are underdeveloped. Because decision-making requires a considerable investment of effort, individuals will be more prone to make financial decisions automatically. In these circumstances, consumers will rely primarily on System 1, which is based on feelings instead of rational thought, since this path will be much easier and less cognitively demanding.

Conclusion

Financial decision-making is a very complex notion. From an evolutionary perspective, it seems likely that multiple motivations would affect any given behaviour. In other words, there is a wide range of shifting factors that might have significant impact on financial decisions and behaviour. Many believe that financial education may address poor individual financial performance. However, the question is whether financial literacy is enough for an individual to accomplish responsible financial and consumption behaviour.

In order to understand and measure consumption decision-making and financial behaviour of consumers, we defined the concept of "responsible financial consumption behaviour" and developed a corresponding index. In terms of sound financial consumption behaviour, individuals want to be able to fulfil their needs and wants to the greatest extent possible. Consumers want enough money to buy the things they like, pay all their financial liabilities on time, live in accordance with their financial capacity, anticipate and plan for their future income and expenses, achieve their financial goals, save for emergencies and so forth. In order to accomplish their countless goals without endangering their financial stability, however, individuals should engage in responsible financial consumption behaviour.

The findings of this study imply that knowledge by itself is not enough; the right attitudes make the actual difference. Therefore, they are of special interest to educators and educational policy-makers who may wish to create attitude-focused workshops where individuals can develop positive and responsible attitudes towards consumption and personal finance management.

One of the main limitations of this study is its reliance on self-reported data. Some participants in the relevant survey might have given false or socially desirable answers. However, this issue is present in every study using self-reported data.

This paper makes a significant contribution to financial planners and counsellors as well. The index of responsible financial consumption behaviour possesses desirable psychometric characteristics. It provides practitioners a reliable and valid tool for measuring the level of individuals' responsibility as they make consumption-related financial decisions.

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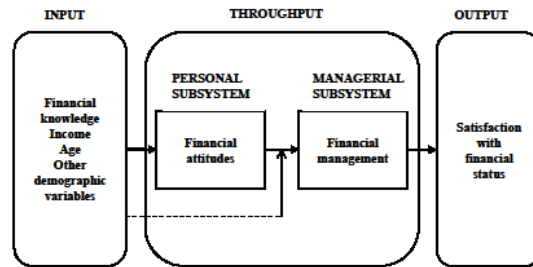
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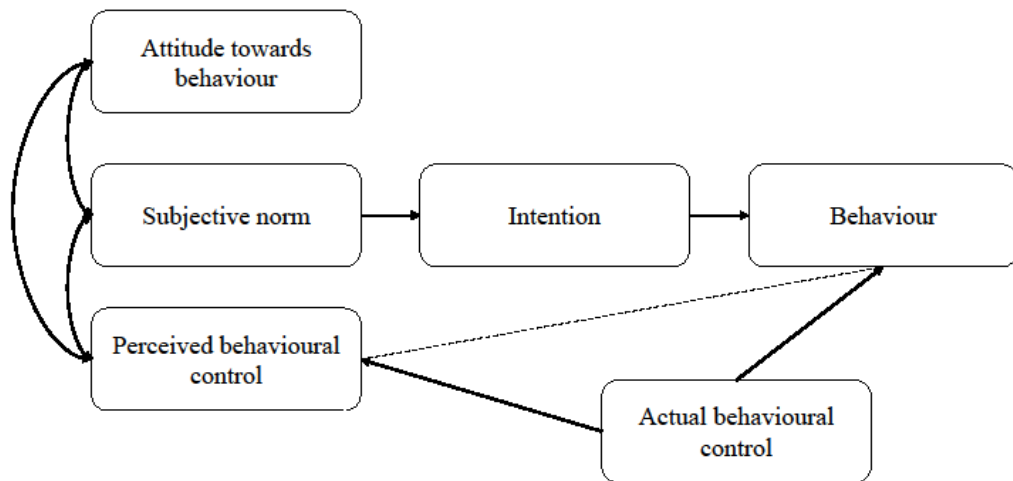
		Factor Loadings	AVE	CR
Attitudes	Att1	0,785	0,552	0,786
	Att2	0,651		
	Att3	0,785		
Control	Contr1	0,761	0,663	0,855
	Contr2	0,818		
	Contr3	0,861		

Hypothesis	Relation direction	Standard Estimate	Standard Error	T statistics	Judgment
H1	Attitudes-> RFCB	0,653	0,098	6,534***	YES
H2	FL ->RFCB	0,072	0,081	0,832***	YES
H3	Control -> RFCB	0,162	0,118	1,329***	YES
H4	Control ->Attitudes	0,418	0,083	4,795***	YES
H5	FL -> Attitudes	0,149	0,092	1,529***	YES
*** p<0,01; ** p<0,05; * p<0,1					

Hypothesis	Relation direction	Standard Estimate	Standard Error	T statistics	Judgment
H6	FL -> Attitudes	0,149	0,092	1,529***	Partial mediation
	Attitudes ->RFCB	0,653	0,098	6,534***	
	FL ->RFCB	0,072	0,081	0,832***	
*** p<0,01; ** p<0,05; *p<0,1					



Source: Parrotta and Johnson (1998)



Source: Ajzen, 2006

